Reivece papers with AI

probability space

Random Experiment: An experiment is said to be random if its result cannot be determined beforehand.

Sample space: The set Ω of all possible results of a random experiment is called a sample space. tose a coin.

$\Omega = \{H, T\}$

Discrete sample space: A sample space Ω is called discrete if it is either finite or countable. A random experiment is called finite if or discrete if its sample space is finite.

Reivew papers with AI

probability space

Random Experiment: An experiment is said to be random if its result cannot be determined beforehand.

Sample space: The set Ω of all possible results of a random experiment is called a sample space. toss a coin.

$\Omega = \{H, T\}$

Discrete sample space: A sample space Ω is called discrete if it is either finite or countable. A random experiment is called finite if or discrete if its sample space is finite.

ASIAN UNIVERSITY OF BANGLADESH

Additional Answer Sheet

ID NO. ______ Program ______

Signature of Invigilator : ______ Semester ______ Course Code ______

Mutually Exclusive Event: Two events A and B are said to be mutually exclusive if $\boxed{A \cap B = \phi}$.

Sigma field / σ-field / σ algebra: Let $\Omega \neq \phi$ A collection F of subsets of Ω is called a σ-algebra over Ω.

(i) If $\Omega \in F$.

(ii) If $A \in F$, then $A^c \in F$.

(iii) If $A_1, A_2 \dots \in F$, then $\bigcup_{i=1}^{\alpha} A_i \in F$.

F = called events / collection of event

→ $A_1 \cup A_2 \cup A_3 \cup \dots \in F$

→ is any collection F of subset of Omega satisfying.

Ex: Toss a coin $\Omega = \{H, T\}$ & A σ-feild $F = \{\phi, \{H\}, \{T\}, \{H, T\}\}$ check it is it a σ-field?

Ans:

→ $\Omega \in F$

→ $\Omega \in F$ & $\Omega^c \in$

→ $\{H\} \cup \{T\} \in F$ & $\{T\}, \{H\} \in F$.

Trivial σ-feild: The trivial σ-~~feild~~ field is the smallest possible σ-field on a sample space. Ex: $F = \{\phi, \Omega\}$

Events: An event is a subset of the sample space Ω.

Operations on Events

→ Union $(A \cup B)$: A or B occurs

→ Intersection $(A \cap B)$: Both A and B occur

→ complement (A^c): A does not occur.

→ Difference $(A \setminus B)$: A occurs bu B does n

ASIAN UNIVERSITY OF BANGLADESH

Additional Answer Sheet

ID NO. ________ Program ________

Signature of Invigilator : ________ Semester ________ Course Code ________

Ex: consider the experiment of tossing a coin once. Find collection is σ-field or not:

Soln: The sample space is $\Omega = \{H, T\}$.

The collection $F = \{\phi, \{H\}, \{T\}, \Omega\}$ satisfies all the properties of a σ-field on Ω.

Ex: $\Omega = \mathbb{R} = \{x \mid -\alpha < x < \alpha\}$

$F_0 = \{\phi, \Omega\}$ — trivial σ-field / smallest

$F_1 = \{\phi, (-\alpha, 0), [0, \alpha), \Omega\}$

$[a,b], (-\alpha, a), [a, \alpha), (a,b),$

$F' = \{\phi, \{a\}, (a,\alpha), [a,b], [a,b), \Omega\}$

$F = \{\phi, \{a\}, (a,\alpha), [a,\alpha), (-\alpha, a), (-\alpha, a],$

$(a,b), [a,b), [a,b], (a,b], (-\alpha, b),$

$(-\alpha, b], (b, \alpha),$

Borel σ-field: / $\mathcal{B}$

The Borel σ field, denoted by $\mathcal{B}$, is the σ field generated by all intervals of the form $(-\infty, a]$, $a \in \mathbb{R}$

→ The σ- field generated by the class of intervals $(-\infty, a]$, $a \in \mathbb{R}$ is called the Borel σ- field and is denoted by $\mathcal{B}$

probability:

measurable space: A pair (Ω, F), where Ω is a sample space and F is a σ- field on Ω.

→ it define what outcomes exist and which events we are allowed to measure.

$A \cap B = \phi$

ASIAN UNIVERSITY OF BANGLADESH

Additional Answer Sheet

ID NO.______ Program ______

Signature of Invigilator : ______ Semester ______ Course Code ______

Mutually Exclusive Events:

→ Two events are mutually exclusive if they cannot happen at the same time.

→ If one happens, the other must not happen.

Def: Two event A and B are called mutually exclusive if $A \cap B = \phi$

Ex: let,

A = getting Head $= \{H\}$

B = getting Tail $= T$

$A \cap B \neq \phi$

Relative Frequency: Relative frequency tells us how often an event happens compared to total trials.

→ if an experiment is performed n times and an event A occurs m times then the relative frequency of A is:

$$\text{Relative frequency of } A = \frac{m}{n}.$$

$$R.A = \frac{\text{Number of times event occurs}}{\text{Total number of trials}}$$

<u>Probability space (Ω, F, P)</u>, axiomatic defination of probability

→ Ω → tell possible outcomes.
→ F → Tell allowed events.
→ P → tells chances.

A real-valued function p defined over F satisfying the conditions.

(i) $P(A) \geq 0$ for all $A \in F$ (nonnegative property)

(ii) $P(\Omega) = 1$ (normed property)

(iii) if $A_1, A_2, ..$ are mutually exclusive even in F that is $A_i \cap A_j = \phi$ for all $i \neq j$

= countable addition.

Ex: Let $\Omega = \{1,2,3\}$ with $P(1)=0.2$, $P(2)=0.5$, $P(3)=0.3$ verify whether this defines a probability space.

Soln:

Non negativity: $P(i) \geq 0 \quad \forall i \in \Omega$.

Total probability: $P(\Omega) = 0.5 + 0.3 + 0.2$
$= 1$.

$\therefore$ although all conditions satisfy. Hence This is a valid probability space.

Ex: Let $\Omega = \{1,2,3,4\}$ with $P(i)=k$, $i=1,2,3,4$. find k so that this is a probability space.

Sol: $P(\Omega) = 4k = 1$
$\Rightarrow k = \frac{1}{4}$.

k

ASIAN
UNIVERSITY OF
BANGLADESH

Additional Answer Sheet

ID NO.________ Program ________

Signature of Invigilator : ________ Semester________ Course Code ________

Ex: Let $\Omega = \{H, T\}$, $F = \{\phi, \Omega\}$.
Define $P(\Omega) = 1$. verify whether this is a probability space.

Soln: $P(\phi) = 0$
$P(\Omega) = 1$

Theorem (properties) probability space.

1. $P(\phi) = 0$: ~~the~~ The probability of an impossible event is zero.

2. ~~P(A^c~~ complement Rule: $P(A^c) = 1 - P(A)$
probability of "not A" equals.

Ex. Let $\Omega = \{H, T\}$, $F = \{\phi, \Omega\}$.
Define $P(\Omega) = 1$. verify whether this is a probability space.

Soln: $P(\phi) = 0$
$P(\Omega) = 1$

Theorem (properties) probability space.

1. $P(\phi) = 0$: ~~the~~ The probability of an impossible event is zero.

2. ~~P(A^c~~ complement Rule. $P(A^c) = 1 - P(A)$
probability of "not A". equals.

Additional Answer Sheet

ASIAN UNIVERSITY OF BANGLADESH

ID NO. ____ Program ____

Signature of Invigilator : ____ Semester ____ Course Code ____

conditional probability

Let (Ω, F, P) be a probability space if $A, B \in F$ with $P(A) > 0$ then the probability of the event B under the condition A is defined as $P(B/A) = \frac{P(A \cap B)}{P(A)}$

→ if A and B are two events with $P(A) > 0$ then the conditional probability of B given A is:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

conditional probabilities (Independent Events):

Two events A and B are said to be independent if and ~~of~~ if: $P(A \cap B) = P(A) \cdot P(B)$.

Definition: pairwise Independent Events: A family of events $\{A_i : i \in I\}$ is said to be pairwise (2x2) independent if: $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$

Definition: mutually Independent Events: A sequence of events $\{A_i, i = 1, 2, \dots\}$ is said to be mutually indenpent if and only if for any finite sub collection $A_1, A_2, \dots, A_K$

$P(A_1 \cap A_2 \dots \cap A_K) = P(A_1)P(A_2) \dots P(K)$ for all K.

Example: Let $\Omega = \{1, 2, 3, 4\}$ Let $A = \{1, 2\}$; $B = \{1, 3\}$, $C = \{1, 4\}$ asume that $P(\{\omega\}) = \frac{1}{4}$, $\omega = 1, 2, 3, 4$ is satisfies the pairwise independent and mutually independent.

ASIAN UNIVERSITY OF BANGLADESH

Additional Answer Sheet

ID NO. ________ Program ________

Signature of Invigilator : ________ Semester ________ Course Code ________

Ans: $P(A) = \frac{2}{4} = \frac{1}{2}$; $P(C) = \frac{2}{4} = \frac{1}{2}$

$P(B) = \frac{2}{4} = \frac{1}{2}$

$P(A \cap B) = \frac{1}{4}$, $P(B \cap C) = \frac{1}{4}$

$P(A \cap C) = \frac{1}{4}$

$P(A \cap B \cap C) = \frac{1}{4}$

Here,

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

$\Rightarrow \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$

$\neq \frac{1}{8}$

Hence this is not satisfy mutually independent conditions.

$P(A \cap B) = P(A) \cdot P(B)$

$\Rightarrow \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$

$\Rightarrow \frac{1}{4} = \frac{1}{4}$

and. $P(B \cap C) = P(B) \cdot P(C)$

$$= \frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{1}{4} = P(B \cap C)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$= \frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{1}{4} = P.$$

~~the pair~~

So, A, B C are pairwise Independent event ~~no~~ But they are mutually Independent events.

Measurable

→ Measurable মানে হলো যে তথ্য sigma field জানে, ফাংশান তার বাইরে কিছু তৈরি করতে পারে

Let $X : \Omega \to \mathbb{R}$ the x will be measurable

if $\{\omega : X(\omega) \leq \mathbb{R}\} \in F$,

ω - হলো sample space Ω এর একটি individual outcom.

$$^{n}C_{r} = \frac{n!}{(n-r)!\, r!}$$

ASIAN UNIVERSITY OF BANGLADESH

Additional Answer Sheet

ID NO. ______ Program ______

Signature of Invigilator : ______ Semester ______ Course Code ______

Chapter-2:

Random variable

~~R~~ Defination: A random variable is a rule (function) that assigns a number to each outcome of a random experiment.

→ A random variable is a function $\boxed{X : \Omega \to \mathbb{R}}$ that assigns a real number to every outcome in the sample space

→ Let (Ω, F, P) be a probability space A real valued function $X : \Omega \to \mathbb{R}$ is said to a random variable, if $\boxed{X^{-1}\{(-\alpha, x]\} \in F}$ for all $x \in \mathbb{R}$.

$$X : \Omega \to \mathbb{R}$$

Ex: Let $\Omega = \{a,b,c\}$ $F = \{\phi, \{a\}, \{b,c\}, \Omega\}$

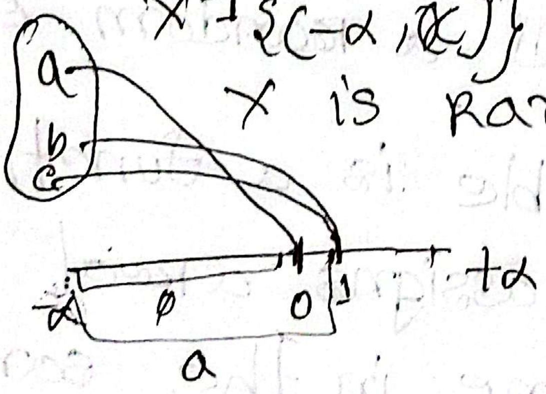
Define $X : \Omega \to R$ such that

$$X(\omega) = \begin{cases} 0 ; & \omega = \{a\} \\ 1 & \omega = \{b\} \text{ or } \omega = \{c\} \end{cases}$$

Soln:

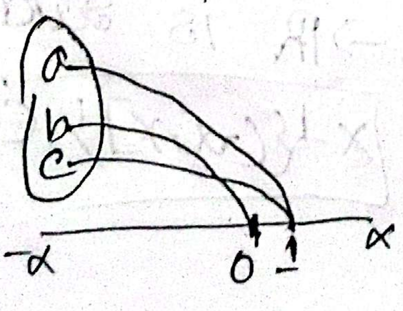
$$X^{-1}\{(-\alpha, x]\} = \begin{cases} \phi & \text{if } x < 0 \\ \{a\} & \text{if } 0 \le x < 1 \\ \Omega & \text{is } x \ge 1 \to (-\alpha, x) \end{cases}$$

While the mapping X^{-1} Here $X^{-1}\{(-\alpha, x]\} \in F$ Hence X is Random variey



Ex: If $X(\omega) = \begin{cases} 0 & \text{if } \omega = b \\ 1 & \text{if } \omega = a \text{ or } \omega = c \end{cases}$

$\Rightarrow$



$$X^{-1}\{(-\alpha, x]\} = \begin{cases} \phi & x < 0 \\ b & 0 \le x < 1 \\ abc\,\Omega & x \le \end{cases}$$

$X^{-1}((-\alpha, x]) = b \notin F.$ so this no powerset $= 2^n$.

ASIAN UNIVERSITY OF BANGLADESH

Additional Answer Sheet

ID NO. ______ Program ______

Signature of Invigilator : ______ Semester ______ Course Code ______

Distribution function/CDF

Defn: A real valued function $F(x)$ satisfying

(i) $0 \leq F(x) \leq 1$ $\forall_x$ (For all) $-\alpha < x < \alpha$

(2) monotonically increasing function in x i.e; if $x_1 < x_2$. then $F(x_1) \leq F(x_2)$

(3) $\lim_{x \to \alpha} F(x) = 0$ & $\lim_{x \to +\alpha} F(x) = 1$

(4) F is (Right) continuous function.

if satisfies all 4 conditions.

Then the function, the real valued function F that is called a distribution function

$$Y^{-1}(-\infty, x] = \begin{cases} \phi & \text{if } x < 0 \\ \{s\} & \text{if } 0 \le x < 1 \\ S & \text{if } x \ge 1 \end{cases}$$

hence $\{s\} \notin F$ so Y is not

the

Distribution function

probabilty distribution (sample, samples probability).

probability mass function: (pmf) : Let X be a distribute Random variable such that $P(X = x) = P_i$ then P_i is said to be probability mass function if it satisfy following condition

(i) $P_i \ge 0$ (ii) $\Sigma P_i = 1$ (sum of all probability)

→ used for Discrete Random variable

Ex: let toss a coin 3 times. then possible sample sapace.

$S \times (S) = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$

So, probility

No. of H	0	1	2	3
P	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

→ probability mas fn

$P(X) = P(X=0) + P(X=1) + P(X=2) + P(X=3) = 1$.

Distribution function:

Let X be a discrete random variable then its discrete distribution function ~~cdf~~ or cumulative ~~function~~ (distribution function (cdf) is defined as

$f(x) = \Sigma P_i = P(X \le x_i)$

Ex:

$$D.F. = \begin{cases} \frac{1}{8} & \text{if } x < 0 \\ \frac{1}{2}\ \frac{4}{8} & \text{if } x < 1 \\ \frac{7}{8} & \text{if } x < 2 \\ 1 & \text{if } x \geq 3 \end{cases}$$

Q(ii).

properties of CDF

(i) $0 \leq F(x) \leq 1 \quad \forall x \quad -\infty < x < \infty$.

(ii) monotonically increasing function in x i.e if $x_1 < x_2$ then $F(x_1) \leq F(x_2)$.

(iii) $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$

(iv) F is right continuous :

PDF (probability Density Function)

→ used for continuous Random vari

→ probability between a and b:

$$P(a<x<b) = \int_a^b f(x)\,dx.$$

Ex: $f(x) = 2x \;;\; 0 < x < 1$

Now find $P(0<x<0.5) = \int_0^{0.5} f(x)\,dx$

$$= \int_0^{0.5} 2x\,dx$$

→ ~~f(x)~~

~~fx~~

$$= \cancel{2}\left[x^2\right]_0^{0.5}$$

$$= (0.5)^2$$

$$= 0.25$$

$\boxed{\begin{array}{l} \rightarrow f(x) \geq 0 \\ \rightarrow \text{Total area} = 1 \text{ so } \int_{-\infty}^{\infty} f(x)\,dx = 1. \end{array}}$

CDF (cumulative Distribution Function)

→ works for Both discrete and continuous.

→ $F(x) = P(X \leq x)$

→ It means probability accumulated up to x.

Ex: (Discret. case) in 3 coin tos

X (N. Head)	$P(x)$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$3/8$
3	$1/8$.

~~for x<0~~

$$CD = \begin{cases} 0 & \text{if } x < 0 \\ 1/8 & \text{if } 0 \leq x \leq 1 \\ 4/8 & \text{if } x \leq 1 \\ 7/8 & \text{if } x \leq 2 \\ 1 & \text{if } x \leq 3 \end{cases}$$

$\rightarrow \quad x \geq 3; \; F_X(x) = P(X=1) + P(X=2) + P(X=3)$

$= 0.2 + 0.5 + 0.3$

$= 1.$

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2 & 1 \leq x < 2 \\ 0.7 & 2 \leq x < 3 \\ 1 & x \geq 3. \end{cases}$$

Absolute continuous Random variables:

$\rightarrow$ Let X be a real random variable defined over the probability space $(\Omega, \mathcal{F}, P)$.

It is said that X is absolutely continuous if and only if there exists a nonnegative and integrable real function f_X such that for all $x \in \mathbb{R}$ it is satisfied that:

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

condition:

Chapter-3

Uniform distribution: A continuous random variables X is said to have a ~~random~~ uniform distribution on the interval [a, b] if its probability density function (PDF) is

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0 & \text{otherwise.} \end{cases}$$

Uniform Random variable: A discrete random variable X is said to have a Discrete uniform Distribution if it takes a finite number of values and each values has equal probability.

If X can take n values then.

$$P(X = x_i) = \frac{1}{n}$$

for each possible value x_i.

→ means all outcomes are equally likely.

Ex: when a fair die is thrown

$\Omega = \{1, 2, 3, 4, 5, 6\}$.

each o/p has probability: $P(X = x) = \frac{1}{6}$

so ~~this~~ this is a discrete uniform Distribution.

~~Also~~ if X takes values $a_1, a_2, a \dots a_n$

$$E[X] = \frac{a_1 + a_2 + \dots + a_n}{n}$$

special case (1 to n)

if $X = \{1, 2, 3, \dots n\}$, $E[X] = \frac{n+1}{2}$

$var[X] = \frac{n^2 - 1}{12}$.

Binomial / Bernoulli distribution:

$$P(X = r) = {}^{n}C_{r}\, p^{r}\, q^{n-r}$$

$q^{n-r} \rightarrow$ failure. $\quad p^{r} \rightarrow$ success

n ~~of tir~~

for $x = 0, 1, \dots n$.

$r \rightarrow$ কতবারের o/p হওয়ার

n = Totall number

→ exactly point থাকলে

Mean $E[X] = np$

$Var(X) = np(1-p) = npq$

A random variable X follows a Binomial distribution if:

→ There are n independent trials.

→ Each trial has only two outcom

→ probability of success p remains constant.

→

Bernoulli distribution: A random variable X follows a Bernoulli distribution if it has only two possi outcomes.

$$X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1- \end{cases}$$

wher $0 \leq p < 1$

It represents a single trial with
→ success (1)
→ failure (0)

PMF:

$$P(X=x) = p^{x}(1-p)^{1-x}$$

mean and variance. $E[X] = p$

$$Var(X) = p(1-p)$$

→ Ex toss a coin once

if probability of head = 0.6.

$$P(x=1) = (0.6)^{1}(1-0.6)^{1-1}$$
$$= 0.6.$$

$$P(X=0) = (0.6)^{0}(1-0.6)^{1-0}$$
$$= 0.4.$$

poisson distribution

poisson Distribution: A discrete random variable X is said to follow a poisson distribution with parameter $\lambda > 0$ if its probability mass function (PMF) is.

$$P(X=k) = \frac{e^{-\lambda}\lambda^{k}}{k!} \quad ; \quad k = 0, 1, 2, 3 \ldots$$

Exponential Distribution

→ A continuous random variable X is said to have an Exponential distribution with parameter $\lambda > 0$ is its probability density function (PDF) is:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

uses for find the weight of the curve at a point

$$P(x \leq 2) = P(x=0) + P(x=1) + P(x=2)$$

range 2ता

$$= \lambda e^{-\lambda 0} + \lambda \cdot e^{-\lambda} + \lambda \cdot e^{-\lambda 2}$$

$f(x) = 1 - e^{\lambda x}$

$$= 3 + 3e^{-3} + 3e^{-6}.$$

$$\text{mean var} = \frac{1}{\lambda} = \frac{\text{Time Inten}}{\text{Average number of arrivals}}$$

Geometric

→ A discrete random variable X is said to follow a geometric distribution with parameter p is

→ we repeat independent trials.

→ Each trial has probability of success p

→ x = number of trials needed to get the first success.

It is denoted by

$$x \sim Geometric(p)$$

$$P(X=k) = (1-p)^{k-1} \cdot p$$

$$E[X] = E[X] = \frac{1}{p}$$

$$Var(X) = \frac{1-p}{p^2}$$

→ have memoryless property

→ only discrete values.

Ex: A coin has probability of Head $p = 0.3$.
Find probability that first head appears on 4th toss.

$$P(X=4) = (1-0.3)^{4-1} \cdot 0.3$$

$$= (0.7)^3 (0.3)$$

$$= 0.1029$$

chapter-4

uniform distribution:

The interval [a,b] is its probab[ility] density function (PDF) is

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0 & \text{otherwise.} \end{cases}$$

Mean:

$$E(x) = \int_{-\alpha}^{\alpha} x\, f(x)dx$$

$$= \int_a^b \frac{1}{b-a} x\, dx$$

$$= \frac{1}{b-a} \left[\frac{x^2}{2}\right]_a^b$$

$$= \frac{1}{b-a} \left[\frac{b^2}{2} - \frac{a^2}{2}\right]$$

$$= \frac{1}{b-a} \frac{(b+a)(b-a)}{2}$$

$$= \frac{b+a}{2}.$$

<u>variance</u> $\text{var}(x) = E(x^2) - (E(x))^2$

$$E[X] = \int_{-\alpha}^{\alpha} x^2 f(x)\,dx.$$

$$= \int_a^b \frac{1}{b-a} x^2\,dx.$$

$$= \frac{1}{b-a}\int_a^b x^2\,dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$

$$= \frac{1}{b-a}\left[\frac{b^3-a^3}{3}\right]$$

$$= \frac{1}{b-a}\cdot\frac{(b-a)(a^2+ab+b^2}{3}$$

$$= \frac{b^2+ab+b2}{3}$$

$$\text{var} = E(x^2) - (E(x))^2$$

$$= \frac{b^2+ab+a^2}{3} - \frac{a^2+2ab+b^2}{4}$$

$$= \frac{4b^2+4ab+4a^2 - 3a^2 + 6ab + 3b^2}{12}$$

$$= \frac{a^2-2ab+b^2}{12} = \frac{(a-b)^2}{12}.$$

$v = E(x^2) - (E(x))^2$

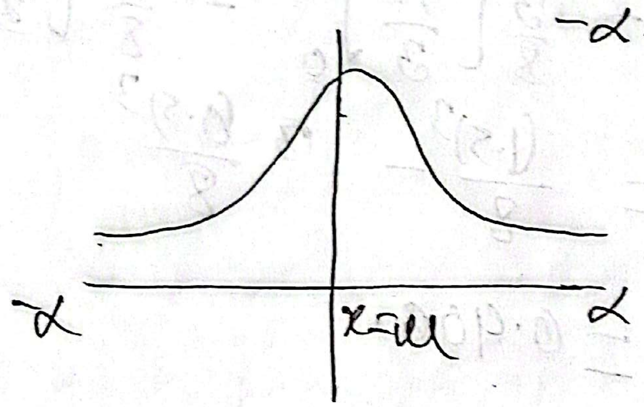
Normal distribution:

A continuous Random variable which has the following p.d.f. (sym

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]; \quad x \in R$$

denonoted $N(\mu, \sigma)$.

$-\alpha < x < \alpha$



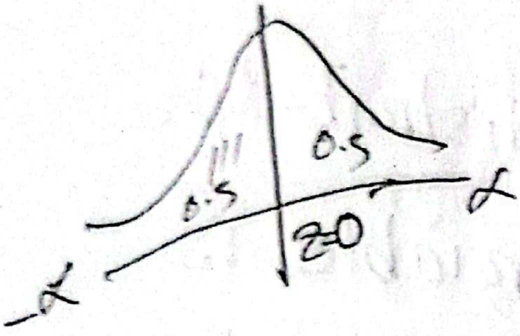
Sandard Normal variable : $\mu = 0$, $\sigma = 1$, then $z = \frac{x-\mu}{\sigma}$.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]}$$

$$= \frac{1}{2\pi} e^{\left(-\frac{1}{2} x^2\right)}$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2}x^2}$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2}z^2}$$ (for ~~nor~~ sandard



<u>**Ex:**</u> $N(1, 4)$

(I) $P(0 \le X < 1$

(II) $P(X^2 > 4)$.

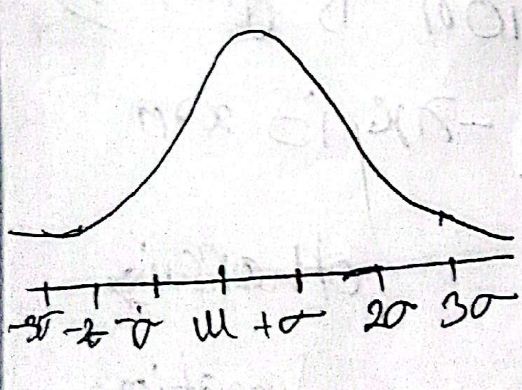
68 + 69

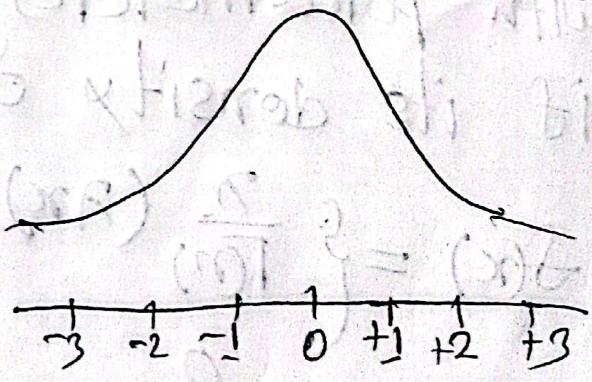
<u>**Soln:**</u>

$$Z = \frac{x-\mu}{\sigma} = \frac{x-1}{4}$$

For 0, $Z_1 = \frac{-1}{4} = -0.25$

For 1 $\quad Z_1 = \frac{0}{4} = 0$





Z-score.

probability Notation: use when dealing with 'conditional chances or likellihoods'.

$$\boxed{P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}}$$

Set Notation: use when dealing with proportion of population or ven diagram problems.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$